SOCIAL MEDIA INFLUENCER SEGMENTATION FOR TARGETED CAMPAIGN

Portfolio link: https://github.com/ssal63014-tech/Social-Media-Influencer-Segmentation-for-Targeted-Campaign

Saldi

PROJECT DESCRIPTION

Objectives

- Develop an influencer segmentation model
- Identify key influencers metrics
- Enhance campaign targeting

Description

Social media influencers play a key role in boosting brand visibility and its engagement. This project leverages machine learning and data analytics to segment influencers based on impact metrics. By analyzing key performance indicators (KPIs), we help brands refine marketing strategies through targeted influencer selection.

Data Science Roles and Responsibilities

Data Scientist: design and implementation, model training and validation, feature engineering.
Data Engineer: data collection, data preprocessing, data storage.
System Architect: infrastructure design, integration, scalability.
Business Analyst: insight interpretation, reporting, strategy formulation.

BUSINESS MODEL

AREAS

Digital Marketing Sector: This is particularly beneficial for brands and marketing agencies looking to optimize influencer collaboration and campaign effectiveness.

EXPECTED BENEFITS

Enhanced campaign targeting, improved marketing ROI and competitive advantages to stay ahead of competitors,

CHALLENGES

Data Privacy and Security,
Data Quality,
Model Accuracy,
Integration.

BENEFIT TARGETS

Marketing Teams,
Brands and Agencies,
Influencers,
Consumers.

01 - DATA COLLECTION

Many social media platforms offer APIs that enable developers to access data. Data also can be collected through collaboration with influencer agencies or third-party influencer marketing platforms.

02 - REQUIRED STORAGE

Cloud platforms like Google Cloud Platform (GCP) or AWS are ideal for dataset due to their scalability and integration capabilities. It is well-suited for storing structured and unstructured data.

03 - SOFTWARE AND TOOLS

To process the dataset, R is the convenient software. packages like dplyr and tidyverse can be used for data manipulation and analysis. Also use cluster, caret, and rpart for data modelling, while ggplot2, pheatmap, rpart.plot, factoextra for visualizing findings.

DATA CHARACTERISTICS

VOLUME

it consists of 5000 rows and 23 columns, which is moderate size but could grow larger with additional campaign or influencer data.

VARIETY

The data contains a mix of numerical (e.g., follower count, engagement rate) and categorical (e.g., campaign type, influencer tier) variables.

```
'data.frame':   5000 obs. of  23 variables:
 $ campaign_id                  : int  1 2 3 4 5 6 7 8 9 10 ...
 $ cmp_influencer_id            : int  272 367 692 888 814 856 2 533 749 974 ...
 $ cmp_campaign_type            : chr  "Product Launch" "Event" "Product Launch" "Promotion" ...
 $ cmp_campaign_duration_days   : int  56 28 16 51 23 42 25 10 40 23 ...
 $ cmp_platform_used            : chr  "YouTube" "Instagram" "TikTok" "TikTok" ...
 $ cmp_content_type_in_campaign : chr  "videos" "reels" "images" "images" ...
 $ cmp_number_of_sponsored_posts: int  6 3 4 5 1 8 9 6 6 5 ...
 $ cmp_total_engagement         : int  90097 104883 209168 524855 272511 288062 157995 365043 251887 596095 ...
 $ cmp_campaign_cost            : int  3008 1612 3313 38615 42399 21738 2021 4048 40149 91370 ...
 $ influencer_id                : int  272 367 692 888 814 856 2 533 749 974 ...
 $ inf_follower_count           : int  61411 62707 11519 293504 377246 396727 19217 96962 534579 2264277 ...
 $ inf_engagement_rate          : num  0.0545 0.0858 0.099 0.057 0.0266 0.045 0.0955 0.0789 0.0225 0.0388 ...
 $ inf_post_frequency           : int  18 10 13 13 10 16 10 11 14 12 ...
 $ inf_content_type             : chr  "videos" "reels" "images" "images" ...
 $ inf_platform                 : chr  "YouTube" "Instagram" "TikTok" "TikTok" ...
 $ inf_niche_category           : chr  "Beauty" "Fashion" "Food" "Food" ...
 $ inf_audience_age_distribution: chr  "25-34" "35-44" "25-34" "25-34" ...
 $ inf_audience_gender          : chr  "Male" "Male" "Both" "Female" ...
 $ inf_audience_location        : chr  "Lombok" "Lombok" "Makassar" "Bali" ...
 $ inf_audience_interests       : chr  "Tech Enthusiasts" "Fitness Buffs" "Foodies" "Fitness Buffs" ...
 $ inf_influencer_tier          : chr  "Micro" "Micro" "Micro" "Macro" ...
 $ inf_engagement_growth_rate   : num  0.352 0.229 0.372 0.179 0.367 ...
 $ inf_average_reach            : int  197759 293844 76155 601158 370069 108436 64486 200725 427698 155064 ...
```

VELOCITY

The dataset could be frequently updated with new campaign information, such as engagement rates or influencer growth, indicating the potential for real-time or frequent data changes.

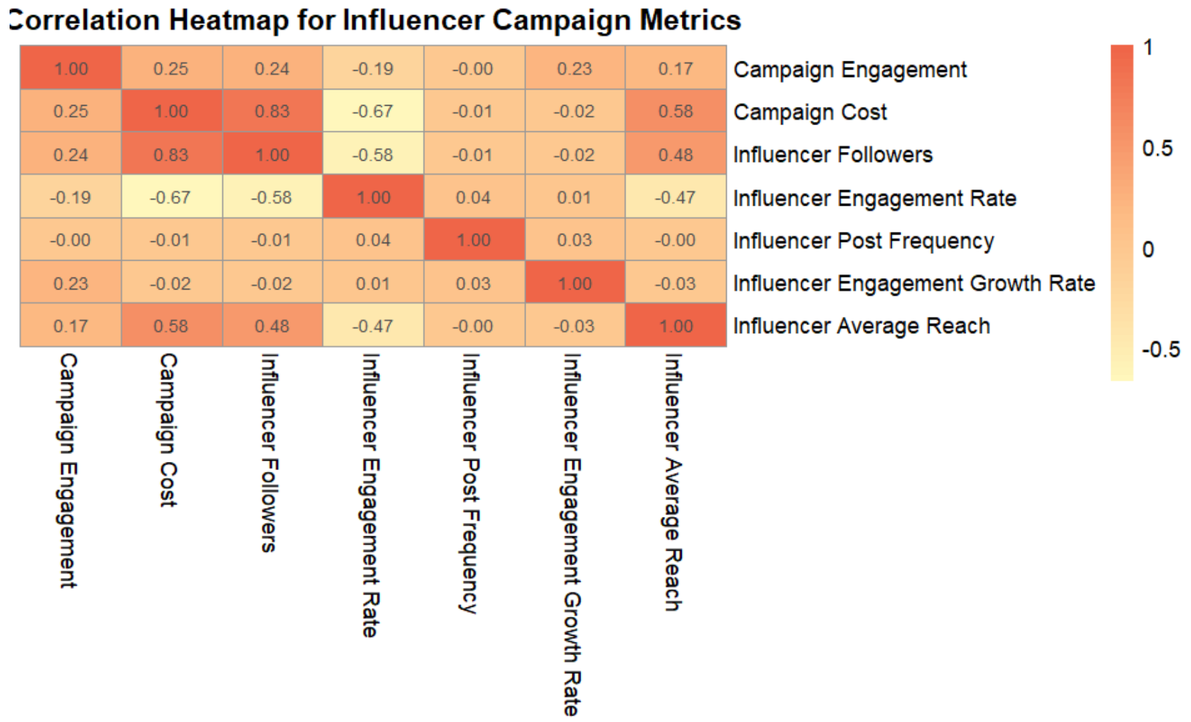
VERACITY

The data quality appears consistent (no missing values), but further validation could be needed to ensure accuracy, especially with influencer metrics such as engagement growth rates and audience data, which may fluctuate over time.

DATA CORRELATION

The strong positive correlation (0.83) between Campaign Cost and Influencer Followers suggests that higher costs are typically associated with influencers with a larger follower base. This implies that brands tend to invest more in influencers with a greater reach, likely due to the perceived potential for higher visibility and broader audience exposure.

There is a correlation of 0.23 between influencer engagement growth rate and campaign engagement, indicating that influencers experiencing increased engagement over time may contribute to higher campaign engagement. However, this effect is relatively weak, suggesting that while growing engagement is a positive indicator, it is not the sole predictor of campaign success.



DATA CLUSTERING

1

CLUSTER 1

influencers with the highest average follower, engagement, and cost. This suggests that these influencers are likely larger in scale, resulting in high engagement but at a higher cost.

2

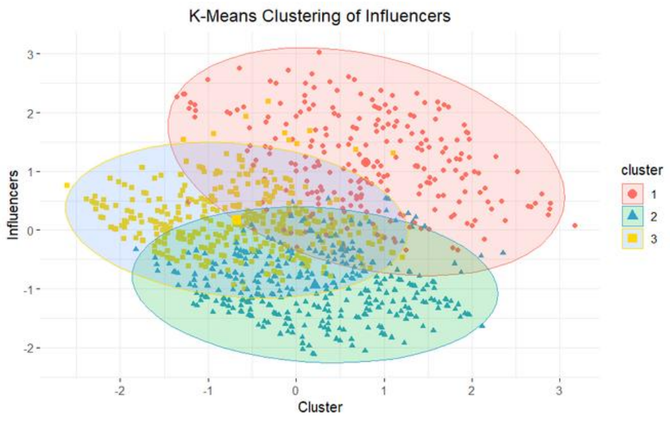
CLUSTER 2

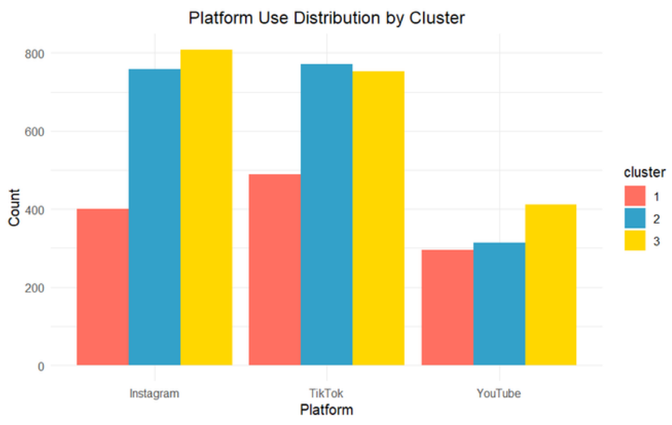
Influencers in the middle range who still contribute to effective campaigns.

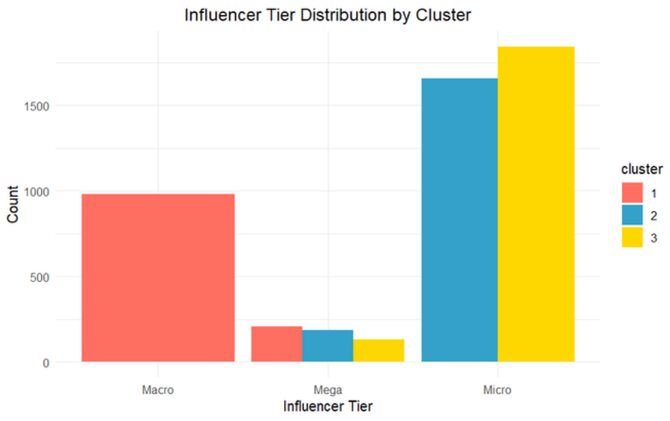
3

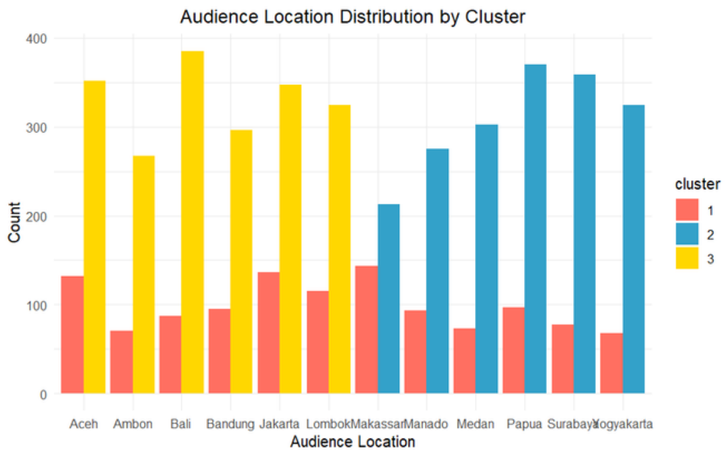
CLUSTER 3

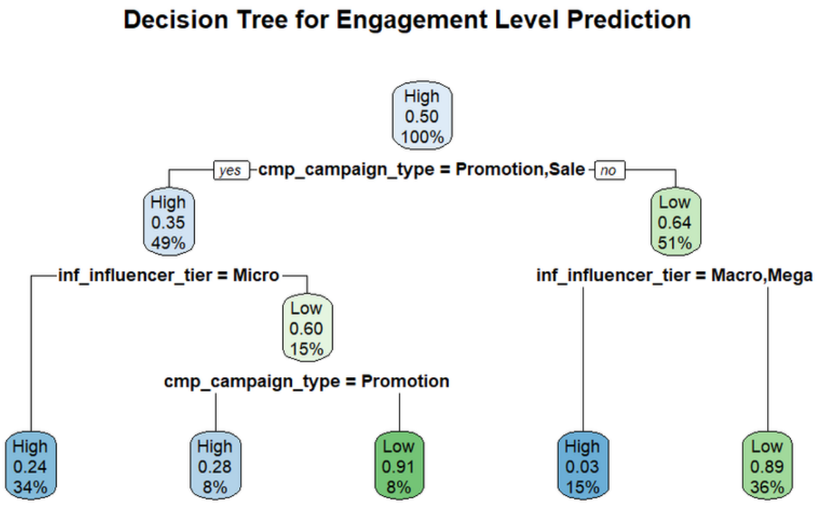
influencers who still contribute to effective campaigns at a relatively lower cost.

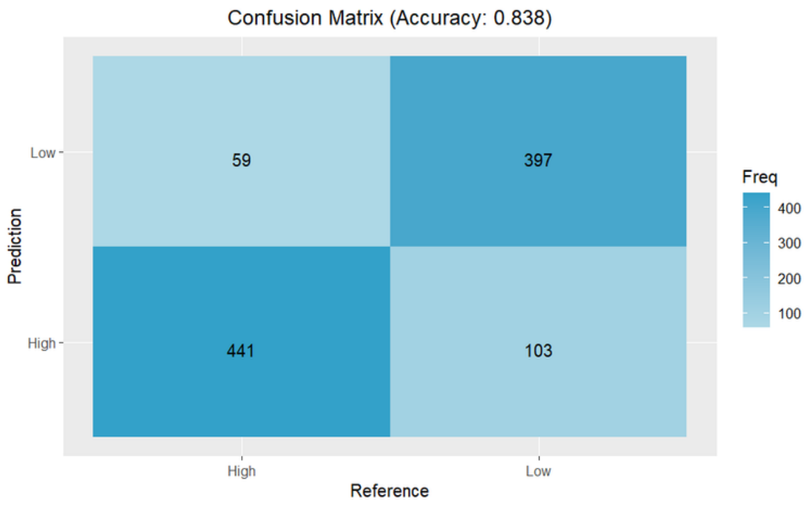












DECISION TREE MODEL

From the picture provided, it aims to predict campaign success based on influencer characteristics and campaign features, specifically the engagement level, which should be high or low.

In business interpretation, understanding the prediction case can help companies allocate their marketing budgets more effectively. This could mean shifting investment away from high-cost Macro influencers in favor of more cost-effective micro influencers who can drive better engagement for certain campaigns.

STANDARD DATA REGULATION

Access is restricted to authorized personnel, with security measures like encryption and anonymization in place to protect sensitive information from influencer and campaign. Influencer details are anonymized to ensure confidentiality, and comply with GDPR for legal standards.

DATA GOVERNANCE

Standard approach followed is CRISP-DM framework that involving six stages of data science process.

1. Aligning business objectives
2. Assessing data quality
3. Cleaning and selecting features
4. Applying correlation check, K-means clustering and decision tree.
5. Evaluating model perfomance
6. Deploying actionable marketing insight into visualization.

DATA SCIENCE PROCESS

Data is systematically organized to maintain the accuracy and readability, version-controlled to track changes, and regularly backed up to prevent loss. Ethical practices include transparency in data use, informing influencers through their agency, and using data solely to enhance marketing strategies, ensuring no harm to influencers or audiences.

DATA MANAGEMENT